

§ 21. The Fundamental Theorem of Algebraic Extensions.

Definition 1. Principal functions and principal factors of the regular well-ordering. –

Let $\mathfrak{N} = \mathfrak{K}_\sigma$ be an algebraic extension of the well-ordered field $\mathfrak{K}_0$, regularly ordered with respect to $\mathfrak{K}_0$, and let $\mathfrak{J} = \mathfrak{J}_\sigma = \{j_\nu\}$ ($0 \leq \nu < \sigma$) be the system of principal elements; let $\mathfrak{K}_\nu$ and $\mathfrak{J}_\nu$ denote, respectively, the principal segment of $\mathfrak{N}$ and the segment of $\mathfrak{J}$ determined by j_ν . The element j_ν is a root of a certain irreducible function $f_\nu(x)$ over $\mathfrak{K}_0$ (with ε as coefficient of the highest power), and likewise a root of a certain irreducible function $\varphi_\nu(x)$ of the field $\mathfrak{K}_\nu$. This $\varphi_\nu(x)$ may be equal to $f_\nu(x)$, and for $\nu = 0$ is always equal to $f_\nu(x)$; in every case it is a factor of $f_\nu(x)$. We thus arrive at two systems of functions

$$F_\sigma = F = \{f_\nu(x)\}, \quad \Phi_\sigma = \Phi = \{\varphi_\nu(x)\} \quad (0 \leq \nu < \sigma).$$

We call the functions $f_\nu(x)$ the *principal functions*, and the functions $\varphi_\nu(x)$ the *principal factors* of the regular well-ordering, and denote by F_ν and Φ_ν the segments of F and Φ determined by $f_\nu(x)$ and $\varphi_\nu(x)$, respectively ($\nu \geq 1$).¹ – With this terminology the following is immediate.

1. *If all principal functions are normal (with respect to $\mathfrak{K}_0$), then the regularly ordered field $\mathfrak{N} = \mathfrak{K}_\sigma$ is itself normal. It is just large enough to split completely the system F of principal functions; likewise each principal segment $\mathfrak{K}_\nu$ is normal and is just sufficient for the splitting of F_ν . The system F is progressive.*

Indeed, each of the normal functions $f_\rho(x)$ ($0 \leq \rho < \nu$) from F_ν ($0 < \nu \leq \sigma$) has in $\mathfrak{K}_\nu = \mathfrak{K}_0(\mathfrak{J}_\nu)$ a root j_ρ ; hence it must split completely in $\mathfrak{K}_\nu$. Since the field $\mathfrak{K}_\nu$ arises from $\mathfrak{K}_0$ solely by adjoining roots of functions from F_ν , it is just sufficient for the splitting of F_ν , and is therefore normal. Moreover, if $\nu < \sigma$, the field $\mathfrak{K}_\nu$ no longer contains the element j_ν ; thus $f_\nu(x)$ does not split completely in $\mathfrak{K}_\nu$, and so F is a progressive system.

Definition 2. Normal well-ordering. – Let $\mathfrak{N}$ be a normal extension of the well-ordered field $\mathfrak{K}_0$; let $\mathfrak{N}$ be regularly ordered with respect to $\mathfrak{K}_0$, and suppose that all principal functions are normal. Keeping the letters in the same meaning as before, j_ν ($0 \leq \nu < \sigma$) is a root of the irreducible function $\varphi_\nu(x)$ over $\mathfrak{K}_\nu$, and $\varphi_\nu(x)$ is a factor of the function $f_\nu(x)$ irreducible over $\mathfrak{K}_0$. Since $\mathfrak{K}_\nu$ is a well-ordered field, Def. 4 of § 20 assigns a definite order also to the functions inside $\mathfrak{K}_\nu$. This therefore also applies to the irreducible factors that $f_\nu(x)$ has inside $\mathfrak{K}_\nu$. We shall now assume that every function $\varphi_\nu(x)$, unless it is the sole irreducible factor of $f_\nu(x)$ inside $\mathfrak{K}_\nu$, precedes the remaining irreducible factors of $f_\nu(x)$ inside $\mathfrak{K}_\nu$. If this condition is fulfilled, we say that $\mathfrak{N}$ is *normally ordered with respect to $\mathfrak{K}_0$* . Thus normal well-ordering is a special case of regular well-ordering.

Definition 3. Distinguished roots of the principal functions belonging to a normal well-ordering. – Let $\mathfrak{N}$ be a normal and normally ordered extension of the well-ordered field $\mathfrak{K}_0$, and let the symbols used so far keep their meanings. Since the principal function $f_\nu(x)$ is normal with respect to $\mathfrak{K}_0$, the field $\mathfrak{K}_\nu(j_\nu)$ can be obtained by adjoining to $\mathfrak{K}_\nu$ any root of $f_\nu(x)$. All these roots are primitive elements of the field $\mathfrak{K}_\nu(j_\nu)$ over $\mathfrak{K}_\nu$; their degree is equal to the degree of the principal factor $\varphi_\nu(x)$, which in any case is greater than 1. In the case of imperfect fields the roots of $\varphi_\nu(x)$ need not all be distinct. Suppose therefore that $\varphi_\nu(x)$ has r distinct roots $j_\nu, j'_\nu, \dots, j_\nu^{(r-1)}$ and has degree n ; then $n \geq r \geq 1$, $n > 1$. If now i_ν is any root of $f_\nu(x)$ (in the field $\mathfrak{N}$), then we obtain all elements of the field $\mathfrak{K}_\nu(i_\nu)$ by replacing x by i_ν in all integral functions $g(x)$ over $\mathfrak{K}_\nu$ whose degree is smaller than n . Among these functions there are r which, for $x = i_\nu$, become roots of $\varphi_\nu(x)$. By Def. 4 of § 20 these r functions have a definite order within $\mathfrak{K}_\nu$. If $\psi(x)$ is the first among them and if $\psi(i_\nu) = j_\nu$, then we call i_ν a *distinguished root* of $f_\nu(x)$. (We also use this designation when $r = 1$, which can occur only for imperfect fields.) The following holds.

¹To avoid misunderstandings, we add that within F the functions $f_\nu(x)$ have a definite order, indicated by the index. This order of course has nothing to do with the order assigned to functions in $\mathfrak{K}_0$ by § 20, Def. 4.

2. *The principal element j_ν is a distinguished root of $f_\nu(x)$, and among the roots of the principal factor $\varphi_\nu(x)$ there is no other distinguished one.*

Indeed, since the roots $j_\nu, j'_\nu, \dots, j_\nu^{(r-1)}$ of $\varphi_\nu(x)$ do not occur in $\mathfrak{K}_\nu$, the r functions over $\mathfrak{K}_\nu$ that for $x = i_\nu$ become these roots all have degree at least 1. From the fact that the well-ordered field $\mathfrak{K}_\nu$ begins with the elements $0, \varepsilon$, it follows that among all functions over $\mathfrak{K}_\nu$ of degree ≥ 1 , the first is x . Thus, if for i_ν we choose one of the roots of $\varphi_\nu(x)$, then x is the first function over $\mathfrak{K}_\nu$ which for $x = i_\nu$ becomes a root of $\varphi_\nu(x)$. But this root is equal to j_ν if and only if we choose $i_\nu = j_\nu$, proving the assertion. – From 2. one further obtains:

3. *In the case $\varphi_\nu(x) = f_\nu(x)$, hence in particular for $\nu = 0$, j_ν is the unique distinguished root of $f_\nu(x)$.*

We now prove:

4. *Let $\mathfrak{N} = \mathfrak{K}_0(\tilde{\mathfrak{J}})$ be a normal extension of the well-ordered field $\mathfrak{K}_0$. Suppose every element i_ν of the system $\tilde{\mathfrak{J}}_\sigma = \tilde{\mathfrak{J}} = \{i_\nu\}$ ($0 \leq \nu < \sigma$) is a root of a normal function $f_\nu(x)$ over $\mathfrak{K}_0$, and suppose the function system $F_\sigma = F = \{f_\nu(x)\}$ is progressive (or, what here amounts to the same thing, suppose the system $\tilde{\mathfrak{J}}$ is progressive). Then the field $\mathfrak{N}$ can be normally ordered with respect to $\mathfrak{K}_0$ in one and only one way so that F is the system of principal functions and the elements i_ν become distinguished roots of the corresponding principal functions $f_\nu(x)$.*

We first show that two distinct normal orderings of the indicated kind cannot exist. Assume that there are two such normal orderings $\mathfrak{N}'$ and $\mathfrak{N}''$ of the field $\mathfrak{N}$. Denote by $\mathfrak{J}'_\sigma = \mathfrak{J}' = \{j'_\nu\}$, $\{\mathfrak{K}'_\nu\}$, $\Phi'_\sigma = \Phi' = \{\varphi'_\nu(x)\}$ the systems of principal elements, principal segments, and principal factors of $\mathfrak{N}'$, and by $\mathfrak{J}''_\sigma = \mathfrak{J}'' = \{j''_\nu\}$, $\{\mathfrak{K}''_\nu\}$, $\Phi''_\sigma = \Phi'' = \{\varphi''_\nu(x)\}$ the corresponding systems for $\mathfrak{N}''$. Let $\tilde{\mathfrak{J}}_\nu, F_\nu, \mathfrak{J}'_\nu, \mathfrak{J}''_\nu, \Phi'_\nu, \Phi''_\nu$ denote the segments of $\tilde{\mathfrak{J}}, F$, and so on determined by $i_\nu, f_\nu(x)$, and so forth. Then $\mathfrak{K}'_\nu$ and $\mathfrak{K}''_\nu$ are two normal orderings of the same field $\mathfrak{K}_0(\tilde{\mathfrak{J}}_\nu) = \mathfrak{K}_0(\mathfrak{J}'_\nu) = \mathfrak{K}_0(\mathfrak{J}''_\nu)$, which is just sufficient for splitting the system F_ν . Since i_0 is to be a distinguished element of $f_0(x)$, it follows by 3. that $j'_0 = i_0 = j''_0$. If the equality $j'_\nu = j''_\nu$ does not hold for every index, take the least one for which $j'_\nu \neq j''_\nu$. Then $\mathfrak{J}'_\nu = \mathfrak{J}''_\nu$, and since $\mathfrak{J}'_\nu$ and $\mathfrak{J}''_\nu$ are the systems of principal elements of the fields $\mathfrak{K}'_\nu, \mathfrak{K}''_\nu$ regularly ordered with respect to $\mathfrak{K}_0$, we have (also taking order into account) $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$ (§ 20, 10.). It follows further that $\varphi'_\nu(x) = \varphi''_\nu(x)$, because $\varphi'_\nu(x)$ and $\varphi''_\nu(x)$ are the first irreducible factor of $f_\nu(x)$ inside $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$. If, furthermore, inside $\mathfrak{K}'_\nu = \mathfrak{K}''_\nu$, $\psi(x)$ is the first function that for $x = i_\nu$ becomes a root of $\varphi'_\nu(x) = \varphi''_\nu(x)$, then, since i_ν is to be a distinguished root of $f_\nu(x)$, we get $j'_\nu = \psi(i_\nu) = j''_\nu$, contrary to the assumption $j'_\nu \neq j''_\nu$. Hence $\mathfrak{J}' = \mathfrak{J}''$, and from § 20, 10. it follows that $\mathfrak{N}' = \mathfrak{N}''$.

It remains to show that a normal ordering of the required kind exists. – For $\sigma = 1$ we obtain it by ordering $\mathfrak{N}$ primitively with respect to $\mathfrak{K}_0$, with i_0 as principal element. For $\sigma > 1$ the proof is by induction. If σ is not a limit number, let $\mathfrak{K}_{\sigma-1}$ denote the normal ordering of the field $\mathfrak{K}_0(\tilde{\mathfrak{J}}_{\sigma-1})$ for which $F_{\sigma-1}$ is the system of principal functions and every element i_ν ($\nu < \sigma - 1$) is a distinguished root of $f_\nu(x)$. If then, within the well-ordered field $\mathfrak{K}_{\sigma-1}$, $\varphi_{\sigma-1}(x)$ is the first irreducible factor of $f_{\sigma-1}(x)$, and $\psi(x)$ is the first function that for $x = i_{\sigma-1}$ becomes a root of $\varphi_{\sigma-1}(x)$, put $\psi(i_{\sigma-1}) = j_{\sigma-1}$ and order $\mathfrak{N}$ primitively with respect to $\mathfrak{K}_{\sigma-1}$, with $j_{\sigma-1}$ as principal element. This plainly gives the required normal ordering. – If σ is a limit number, then for every $\nu < \sigma$ let $\mathfrak{K}_\nu$ be the normal ordering of the field $\mathfrak{K}_0(\tilde{\mathfrak{J}}_\nu)$ for which F_ν is the system of principal functions, while every element i_μ ($\mu < \nu$) is a distinguished root of $f_\mu(x)$. A simple consideration then shows that each field of the sequence $\{\mathfrak{K}_\nu\}$ ($0 \leq \nu < \sigma$) is a segment of every later one, and that the totality of the elements occurring in these fields yields a well-ordering of $\mathfrak{N}$ satisfying all conditions of the problem.

5. *Let $\mathfrak{K}_0$ and $\bar{\mathfrak{K}}_0$ be similarly isomorphic fields. Let $\mathfrak{N} = \mathfrak{K}_\sigma$ be a normal and normally ordered extension of $\mathfrak{K}_0$, and $\bar{\mathfrak{N}} = \bar{\mathfrak{K}}_\sigma$ such an extension of $\bar{\mathfrak{K}}_0$. Let $F = \{f_\nu(x)\}$ and $\bar{F} = \{\bar{f}_\nu(x)\}$ ($0 \leq \nu < \sigma$) be the corresponding systems of principal functions, and suppose that the similarly isomorphic correspondence π_0 between $\mathfrak{K}_0$ and $\bar{\mathfrak{K}}_0$ assigns the functions $f_\nu(x)$ and $\bar{f}_\nu(x)$ to one*

another. Then $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are similarly isomorphic.

Proof by induction. If $\sigma = 1$, if n is the degree of $f_0(x)$ and $\overline{f}_0(x)$, and if $j_0, \overline{j}_0$ are the principal elements, then every element of $\mathfrak{N}$ or $\overline{\mathfrak{N}}$ is obtained by replacing x by j_0 or $\overline{j}_0$ in all functions $g(x)$ or $\overline{g}(x)$ over $\mathfrak{K}_0$ or $\overline{\mathfrak{K}}_0$ of degree smaller than n . The functions $g(x)$ in $\mathfrak{K}_0$ and the functions $\overline{g}(x)$ in $\overline{\mathfrak{K}}_0$ have definite orders, and since π_0 is a similar correspondence between $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$, it follows at once that the correspondence between the functions over $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ induced by π_0 is also similar. Further, from § 20, Def. 5, we obtain a similar correspondence π_1 between $\mathfrak{N} = \mathfrak{K}_0(j_0)$ and $\overline{\mathfrak{N}} = \overline{\mathfrak{K}}_0(\overline{j}_0)$ extending π_0 , by making the element $\overline{g}(\overline{j}_0)$ of $\overline{\mathfrak{N}}$ correspond to the element $g(j_0)$ of $\mathfrak{N}$, where $\overline{g}(x)$ denotes the function over $\overline{\mathfrak{K}}_0$ corresponding under π_0 to $g(x)$ over $\mathfrak{K}_0$. Since π_0 is moreover an isomorphism, it follows, as in § 8, 1., that π_1 is also an isomorphism. – If $\sigma > 1$ and if $\mathfrak{K}_\nu, \overline{\mathfrak{K}}_\nu$ ($\nu < \sigma$) denote the principal segments of $\mathfrak{N}$ and $\overline{\mathfrak{N}}$, then $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$ are normal and normally ordered extensions of $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ whose principal functions are carried into each other by π_0 . We may therefore regard it as proved that $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$ are similarly isomorphic, i.e. that the unique similar correspondence π_ν between $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$ is also an isomorphism. If $\mu < \nu < \sigma$, then $\mathfrak{K}_\mu$ and $\overline{\mathfrak{K}}_\mu$ are corresponding segments of $\mathfrak{K}_\nu$ and $\overline{\mathfrak{K}}_\nu$, and hence are mapped similarly onto one another by π_ν , just as by π_μ . Thus π_μ is a part of π_ν . If σ is a limit number, so that every element of $\mathfrak{N}$ or $\overline{\mathfrak{N}}$ occurs in some field $\mathfrak{K}_\nu$ or $\overline{\mathfrak{K}}_\nu$, it follows that the correspondences π_ν determine a one-to-one mapping π_σ from $\mathfrak{N}$ onto $\overline{\mathfrak{N}}$. A simple consideration shows that π_σ is similar, because each π_ν is similar; likewise π_σ is an isomorphism, because the π_ν are isomorphisms. – If σ is not a limit number, then first there is the similarly isomorphic correspondence $\pi_{\sigma-1}$ between $\mathfrak{K}_{\sigma-1}$ and $\overline{\mathfrak{K}}_{\sigma-1}$ extending π_0 , and it assigns the principal functions $f_{\sigma-1}(x)$ and $\overline{f}_{\sigma-1}(x)$ to one another. Moreover, $\pi_{\sigma-1}$ transforms the principal factor $\varphi_{\sigma-1}(x)$, the first factor of $f_{\sigma-1}(x)$ inside $\mathfrak{K}_{\sigma-1}$, into the first factor of $\overline{f}_{\sigma-1}(x)$ inside $\overline{\mathfrak{K}}_{\sigma-1}$, namely $\overline{\varphi}_{\sigma-1}(x)$. The fields $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are normal and normally ordered extensions of $\mathfrak{K}_{\sigma-1}$ and $\overline{\mathfrak{K}}_{\sigma-1}$ with only one principal function, $\varphi_{\sigma-1}(x)$ and $\overline{\varphi}_{\sigma-1}(x)$ respectively. Since $\mathfrak{K}_{\sigma-1}$ and $\overline{\mathfrak{K}}_{\sigma-1}$ are mapped similarly and isomorphically by $\pi_{\sigma-1}$ and $\pi_{\sigma-1}$ also carries the principal functions into one another, the existence of the similarly isomorphic correspondence between $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ follows as in the case $\sigma = 1$.

Letting in Theorem 5 $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$ coincide as one and the same well-ordered field, we obtain:

6. If $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are normal and normally ordered extensions of the well-ordered field $\mathfrak{K}_0$ and the same system $\{f_\nu(x)\}$ of principal functions belongs to both, then $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are similarly isomorphic.

7. If $\mathfrak{K}_0$ is a well-ordered field and $F = F_\sigma = \{f_\nu(x)\}$ ($0 \leq \nu < \sigma$) is a progressive system of normal functions over $\mathfrak{K}_0$, then there exist normal and normally ordered extensions of $\mathfrak{K}_0$ with F as their system of principal functions.

Proof by induction. In the case $\sigma = 1$, by § 8, 1. we can determine an extension $\mathfrak{N}$ that is just sufficient for the splitting of $f_0(x)$. It is normal; and since $f_0(x)$ is normal, we have $\mathfrak{N} = \mathfrak{K}_0(j_0)$ if j_0 is any root of $f_0(x)$ in $\mathfrak{N}$. If we order $\mathfrak{N}$ primitively with respect to $\mathfrak{K}_0$ so that j_0 is principal element, then it is immediately clear that $\mathfrak{N}$ satisfies the conditions of the theorem. – If $\sigma > 1$ and is not a limit number, let $\mathfrak{K}_{\sigma-1}$ be a normal and normally ordered extension of $\mathfrak{K}_0$ with principal-function system $F_{\sigma-1} = \{f_\nu(x)\}$ ($0 \leq \nu < \sigma - 1$). Among the irreducible factors that the normal function $f_{\sigma-1}(x)$ has in the well-ordered field $\mathfrak{K}_{\sigma-1}$, let $\varphi_{\sigma-1}(x)$ be the first. Its degree must be greater than 1, because $\mathfrak{K}_{\sigma-1}$ is just sufficient for the splitting of $F_{\sigma-1}$ (Theorem 1) and the normal-function system F is progressive. If one extends $\mathfrak{K}_{\sigma-1}$ by adjoining a root $j_{\sigma-1}$ of $\varphi_{\sigma-1}(x)$ to a field $\mathfrak{N}$ and orders this field primitively with respect to $\mathfrak{K}_{\sigma-1}$ with $j_{\sigma-1}$ as principal element, then $\mathfrak{N}$ plainly satisfies the conditions of Theorem 7. – If σ is a limit number, let $\mathfrak{K}'_\nu$ ($0 < \nu < \sigma$) be a normal and normally ordered extension of $\mathfrak{K}_0$ with principal-function system $F_\nu = \{f_\lambda(x)\}$ ($0 \leq \lambda < \nu$). If $\mu < \nu$, then $\mathfrak{K}'_\nu$ has a principal segment that is normal and normally ordered with respect to $\mathfrak{K}_0$ and has principal-function system F_μ . By 6 this segment must be similarly isomorphic to $\mathfrak{K}'_\mu$. Thus, among the fields of

the sequence $\mathfrak{K}_0, \{\mathfrak{K}'_\nu\}$ ($0 < \nu < \sigma$), each is similarly isomorphic to a segment of every later one. Hence (§ 20, 4.) there is a well-ordered field $\overline{\mathfrak{N}}$ such that $\mathfrak{K}_0$ is similarly isomorphic to a segment $\overline{\mathfrak{K}}_0$, each field $\mathfrak{K}'_\nu$ ($0 < \nu < \sigma$) is similarly isomorphic to a segment $\overline{\mathfrak{K}}_\nu$ of $\overline{\mathfrak{N}}$, and every element of $\overline{\mathfrak{N}}$ occurs in some such segment $\overline{\mathfrak{K}}_\nu$. Since a similarly isomorphic correspondence exists between $\mathfrak{K}_0$ and $\overline{\mathfrak{K}}_0$, it follows that one obtains a field $\mathfrak{N}$ similarly isomorphic to $\overline{\mathfrak{N}}$, and at the same time an extension field of $\mathfrak{K}_0$, by replacing the segment $\overline{\mathfrak{K}}_0$ of $\overline{\mathfrak{N}}$ by the well-ordered system $\mathfrak{K}_0$ and requiring that every composition equation ($a + b = c$, $a \cdot b = d$) in $\overline{\mathfrak{N}}$ become a composition equation in $\mathfrak{N}$, whenever the elements of $\overline{\mathfrak{K}}_0$ that may occur in it are substituted by the corresponding elements of $\mathfrak{K}_0$. By this substitution each segment $\overline{\mathfrak{K}}_\nu$ of $\overline{\mathfrak{N}}$ passes into a segment $\mathfrak{K}_\nu$ of $\mathfrak{N}$ similarly isomorphic to it, and hence also to $\mathfrak{K}'_\nu$, and every element of $\mathfrak{N}$ occurs in such segments $\mathfrak{K}_\nu$. Through the similarly isomorphic correspondence π_ν between $\mathfrak{K}'_\nu$ and $\mathfrak{K}_\nu$, all order relations and equations of one field are carried into correct order relations and equations of the other. Since π_ν does not alter the elements of $\mathfrak{K}_0$, nor therefore the functions of the system F , $\mathfrak{K}_\nu$, like $\mathfrak{K}'_\nu$, is a normal and normally ordered extension of $\mathfrak{K}_0$ with F_ν as principal-function system. From the fact that the field $\mathfrak{N}$ is composed of the elements of the field sequence $\{\mathfrak{K}_\nu\}$, and that each field of this sequence is a segment of every later one and a segment of $\mathfrak{N}$, it is immediately clear that $\mathfrak{N}$ itself is normally ordered with respect to $\mathfrak{K}_0$, and that F is the corresponding system of principal functions.

From what has gone before one easily obtains the fundamental theorem:

8. *If $\mathfrak{K}_0$ is an arbitrary field and S is an arbitrary system of integral functions over $\mathfrak{K}_0$, then there is an extension of $\mathfrak{K}_0$, unique up to equivalence, which is just large enough to split all functions of the system S into linear factors.*

Proof. For the question considered here, the system S may be replaced by any equivalent system (§ 18, 3.); hence it may be assumed that S contains only normal functions (§ 18, 6.). Moreover we may regard S as well-ordered and, if it is not progressive, replace it by an equivalent progressive subsystem (§ 20, 1.). Thus from the outset we may assume that a progressive system of normal functions $F = \{f_\nu(x)\}$ is given. It is further permissible to suppose that the field $\mathfrak{K}_0$ is well-ordered. But then by 7. there is a normal and normally ordered extension $\mathfrak{N}$ of $\mathfrak{K}_0$ with F as principal-function system, and by 1. this extension is just sufficient for the complete splitting of F . – Now suppose that there is a second extension $\overline{\mathfrak{N}}$ of $\mathfrak{K}_0$ which is just sufficient for the complete splitting of F . We must prove the equivalence of the extensions $\mathfrak{N}$ and $\overline{\mathfrak{N}}$. For this purpose we use the principle of choice. Namely, from the roots that the functions $f_\nu(x)$ have in $\overline{\mathfrak{N}}$, select one for each ν , so that we obtain a system $\tilde{\mathfrak{J}} = \{i_\nu\}$ where i_ν is a root of $f_\nu(x)$. Since these are normal functions, $\overline{\mathfrak{N}} = \mathfrak{K}_0(\tilde{\mathfrak{J}})$. By 4. $\overline{\mathfrak{N}}$ can be normally ordered with respect to $\mathfrak{K}_0$ so that F becomes the system of principal functions (and so that the elements i_ν become distinguished roots of the principal functions. This latter point is now of no further importance, but without it Theorem 4 could not have been proved). Thus $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are normal and normally ordered extensions of $\mathfrak{K}_0$ with the same system of principal functions, and there exists between them a similarly isomorphic correspondence π (Theorem 6). Since under this correspondence every element of the common segment $\mathfrak{K}_0$ is assigned to itself, the extensions $\mathfrak{N}$ and $\overline{\mathfrak{N}}$ are equivalent.

By specialization, 8. further gives:

9. *Every field can be extended algebraically to an algebraically closed field, and essentially in only one way. – In this way to every arbitrary field type there corresponds a definite closed one.*

Indeed, if by S we understand the system of all integral functions over $\mathfrak{K}_0$, then an extension field of $\mathfrak{K}_0$ is algebraically closed and algebraic over $\mathfrak{K}_0$ if and only if it is just sufficient for the splitting of S . Hence the first part of 9. follows from 8. – Let any field type be given; let $\mathfrak{K}_0$ and $\mathfrak{K}'_0$ be two fields of this type, and let Ω and Ω' be algebraic and algebraically closed extensions of $\mathfrak{K}_0$ and $\mathfrak{K}'_0$. We must show that Ω and Ω' are of the same type. To this end introduce a system $\mathfrak{S}$ of new elements, which we put in one-to-one correspondence with the elements of Ω' that

do not belong to $\mathfrak{K}'_0$. This assignment, together with an isomorphism π_0 between $\mathfrak{K}_0$ and $\mathfrak{K}'_0$, gives a mapping π_1 between Ω' and $\bar{\Omega} = \mathfrak{K}_0 + \mathfrak{S}$. If we require that under π_1 every composition equation $a' + b' = c'$, $a' \cdot b' = d'$ from Ω' , even when a' and b' do not both belong to $\mathfrak{K}'_0$, shall pass into a composition equation in $\bar{\Omega}$, then $\bar{\Omega}$ becomes a field isomorphic to Ω' , hence algebraically closed, and is an algebraic extension of $\mathfrak{K}_0$. Therefore, by the first part of 9. already proved, Ω and $\bar{\Omega}$ are equivalent extensions of $\mathfrak{K}_0$, and we have an isomorphic correspondence $\bar{\pi}_1$ between Ω and $\bar{\Omega}$ that leaves the elements of $\mathfrak{K}_0$ unchanged. The desired isomorphic correspondence between Ω and Ω' is obtained from $\bar{\pi}_1$ and π_1 . It is an extension of the isomorphism π_0 existing between $\mathfrak{K}_0$ and $\mathfrak{K}'_0$.

§ 22. Transcendental Extensions. Algebraic Dependence Among Elements of an Extension Field.

Every non-algebraic extension of a field will be called transcendental. – In the following investigations, which are meant to lead us to a clear classification of all fields, we fix a field $\mathfrak{K}$ and refer to it expressions such as “extension field,” “transcendental,” and so on, unless something else is expressly stated. If $\mathfrak{S}$ is a system of elements of an extension field Ω , and a is an element of Ω , we call a *algebraically dependent* on $\mathfrak{S}$ (with respect to $\mathfrak{K}$) if a is algebraic over the field $\mathfrak{K}(\mathfrak{S})$. We call a system of elements $\mathfrak{S}'$ algebraically dependent on $\mathfrak{S}$ if every element of $\mathfrak{S}'$ is algebraically dependent on $\mathfrak{S}$.

1. *If the element a is algebraically dependent on the system $\mathfrak{S}$, then there is a finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ on which a is algebraically dependent.*

For our assumption implies that a is a root of an integral function $\varphi(x)$ of the field $\mathfrak{K}(\mathfrak{S})$. Since $\varphi(x)$ has only finitely many coefficients, there is a finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ such that these coefficients already occur in $\mathfrak{K}(\mathfrak{S}')$. Then a is algebraically dependent on $\mathfrak{S}'$.

2. *If $\mathfrak{S}_3$ is algebraically dependent on $\mathfrak{S}_2$, and $\mathfrak{S}_2$ on $\mathfrak{S}_1$, then $\mathfrak{S}_3$ is algebraically dependent on $\mathfrak{S}_1$.*

Indeed, put $\mathfrak{K}(\mathfrak{S}_1) = \mathfrak{K}_1$, $\mathfrak{K}(\mathfrak{S}_2) = \mathfrak{K}_2$. It follows from our assumption that $\mathfrak{K}_1(\mathfrak{S}_2)$ is algebraic over $\mathfrak{K}_1$, and $\mathfrak{K}_2(\mathfrak{S}_3)$ is algebraic over $\mathfrak{K}_2$ (§ 7, 7.). Since $\mathfrak{K}_2$ is a subfield of $\mathfrak{K}_1(\mathfrak{S}_2)$, the elements of $\mathfrak{K}_2(\mathfrak{S}_3)$ are algebraic over $\mathfrak{K}_1(\mathfrak{S}_2)$, hence also over $\mathfrak{K}_1 = \mathfrak{K}(\mathfrak{S}_1)$ (§ 7, 9.). Therefore the elements of $\mathfrak{S}_3$ are algebraic over $\mathfrak{K}(\mathfrak{S}_1)$, as required.

Accordingly, if we call two systems $\mathfrak{S}_1, \mathfrak{S}_2$ *algebraically equivalent* (with respect to $\mathfrak{K}$), or simply equivalent, when each is algebraically dependent on the other, then any two systems equivalent to a third are equivalent to each other, and so on.

We call a system $\mathfrak{S}$ of elements of an extension field *algebraically reducible* (with respect to $\mathfrak{K}$), or simply reducible, if it is equivalent to a proper part of itself or consists of a single element algebraic over $\mathfrak{K}$; otherwise we call $\mathfrak{S}$ (algebraically) *irreducible*. It is immediately clear from this definition that in a reducible system containing more than one element, at least one element occurs which is algebraically dependent on the system of the remaining elements. In conjunction with 1. this gives at once:

3. *Every subsystem of an irreducible system is irreducible.*

4. *Every reducible system contains a finite reducible subsystem.*

Purely transcendental extensions. – We call an extension of a field *purely transcendental* if it can be obtained by adjoining an irreducible system $\mathfrak{S}$. If $\mathfrak{S}$ consists of only finitely many elements $x_1, x_2, \dots, x_n$, then we have an extension of the kind considered at the end of § 5: the field $\mathfrak{K}(\mathfrak{S})$ consists of rational functions of the transcendentals, or indeterminates, $x_1, x_2, \dots, x_n$ with coefficients in $\mathfrak{K}$. If the system $\mathfrak{S}$ is infinite, then $\mathfrak{K}(\mathfrak{S})$ consists of the elements of all fields $\mathfrak{K}(\mathfrak{S}')$, where $\mathfrak{S}'$ ranges over finite subsystems of $\mathfrak{S}$. The most general

purely transcendental extension is therefore obtained by introducing an arbitrary system $\mathfrak{S}$ of infinitely many “indeterminates.” Each finite subsystem $\mathfrak{S}'$ of $\mathfrak{S}$ then gives an extension $\mathfrak{K}(\mathfrak{S}')$ of the kind previously considered, and the totality of all fields $\mathfrak{K}(\mathfrak{S}')$ constitutes a field (§ 2, 2.) which arises from $\mathfrak{K}$ by adjoining the irreducible system $\mathfrak{S}$. – Suppose now we have two extensions $\mathfrak{K}(\mathfrak{S}_1)$, $\mathfrak{K}(\mathfrak{S}_2)$, where $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are irreducible systems and the finite or infinite number of their elements (cardinal number, cardinality) is the same; that is, suppose a one-to-one correspondence φ can be established between their elements. Then there is a definite isomorphism π between $\mathfrak{K}(\mathfrak{S}_1)$ and $\mathfrak{K}(\mathfrak{S}_2)$ under which every element of $\mathfrak{K}$ corresponds to itself and the elements of $\mathfrak{S}_1$ and $\mathfrak{S}_2$ correspond in the same way as under φ . This was shown in § 5 for finite systems, and in the infinite case follows at once from the same result by a simple consideration. Thus:

5. *Two purely transcendental extensions $\mathfrak{K}(\mathfrak{S}_1)$, $\mathfrak{K}(\mathfrak{S}_2)$ of a field $\mathfrak{K}$, in which the irreducible systems $\mathfrak{S}_1$, $\mathfrak{S}_2$ contain equally many elements, are equivalent.*

6. *If an irreducible system $\mathfrak{S}$ becomes reducible upon adding an element a , then a is algebraically dependent on $\mathfrak{S}$.*

For the reducible system $\mathfrak{S} + a$ contains a finite reducible system $\mathfrak{T}$. The element a must occur in $\mathfrak{T}$, since otherwise $\mathfrak{T}$ would be a part of $\mathfrak{S}$ and hence irreducible. Let $\mathfrak{T} = \mathfrak{T}' + a$, and suppose $\mathfrak{T}'$ consists of the elements $x_1, x_2, \dots, x_n$. Since $\mathfrak{T}'$ is irreducible, every field in the sequence

$$\mathfrak{K}, \quad \mathfrak{K}_1 = \mathfrak{K}(x_1), \quad \mathfrak{K}_2 = \mathfrak{K}_1(x_2), \quad \dots, \quad \mathfrak{K}_n = \mathfrak{K}_{n-1}(x_n) = \mathfrak{K}(\mathfrak{T}')$$

is transcendental over the preceding one. If $\mathfrak{K}_n(a)$ were also transcendental over $\mathfrak{K}_n$, then (§ 5, 9.) every element of $\mathfrak{T}$ would be algebraically independent of the system of the remaining elements, so that $\mathfrak{T}$ would be irreducible. Since this is not the case, a is algebraic over $\mathfrak{K}_n = \mathfrak{K}(\mathfrak{T}')$, hence algebraically dependent on $\mathfrak{T}'$ and therefore also on $\mathfrak{S}$.

7. *Let $\mathfrak{S}$ be an irreducible system (with respect to $\mathfrak{K}$), and let a be transcendental over $\mathfrak{K}$ but algebraically dependent on $\mathfrak{S}$. Then $\mathfrak{S}$ contains a definite finite subsystem $\mathfrak{T}$ with the following properties: a is algebraically dependent on $\mathfrak{T}$; every subsystem of $\mathfrak{S}$ on which a is algebraically dependent contains $\mathfrak{T}$; if any element of $\mathfrak{T}$ is replaced by a , then $\mathfrak{S}$ passes into an equivalent irreducible system; none of the other elements of $\mathfrak{S}$ has this property.*

Proof. Let $\mathfrak{S}'$ denote any subsystem of $\mathfrak{S}$ on which a is algebraically dependent (not excluding $\mathfrak{S}$ itself). Then $\mathfrak{S}'$ has finite subsystems on which a is algebraically dependent. Let $\mathfrak{T}$ be such a subsystem with as few elements as possible. Let its elements be $\alpha_1, \alpha_2, \dots, \alpha_m$, and let $\mathfrak{T}_i$ ($i = 1, \dots, m$) denote the system that remains after omitting α_i from $\mathfrak{T}$ (in the case $m = 1$, the empty system). Since a is algebraically dependent on $\mathfrak{T}$ and algebraically independent of each of the systems $\mathfrak{T}_i$, the system $\mathfrak{T} + a$ is reducible, whereas the systems $\mathfrak{T}_i + a$ are irreducible (Theorem 6). This remains true in the case $m = 1$, because a is transcendental over $\mathfrak{K}$. Since the irreducible system $\mathfrak{T}_i + a$ passes into the reducible system $\mathfrak{T} + a$ by adjoining α_i , the element α_i is algebraically dependent on $\mathfrak{T}_i + a$, and the systems $\mathfrak{T}_i + a$ and $\mathfrak{T} + a$ are equivalent. Likewise $\mathfrak{T}$ and $\mathfrak{T} + a$ are equivalent. Thus we see: *The $m + 1$ systems $\mathfrak{T}_i + a$, $\mathfrak{T}$, and $\mathfrak{T} + a$, and therefore also any two of them, are equivalent.*

We now show that every subsystem $\mathfrak{S}''$ of $\mathfrak{S}$ on which a is algebraically dependent must contain the system $\mathfrak{T}$. To this end let $\mathfrak{U}$ be a finite subsystem of $\mathfrak{S}''$ with as few elements as possible among those on which a is algebraically dependent, and prove that $\mathfrak{U}$ and $\mathfrak{T}$ are identical. Suppose $\mathfrak{U}$ consists of the elements $\beta_1, \dots, \beta_n$, and let $\mathfrak{U}_i$ ($i = 1, \dots, n$) be the system left after omitting β_i from $\mathfrak{U}$. Then the systems $\mathfrak{U}_i + a$ and $\mathfrak{U}$ are irreducible and are equivalent to one another and to the reducible system $\mathfrak{U} + a$. – Because $\mathfrak{T}$ and $\mathfrak{U}$ are smallest systems on which a is algebraically dependent, neither of them can be a proper part of the other. If they were different, we could therefore assume that α_1 is not in $\mathfrak{U}$ and β_1 is not in $\mathfrak{T}$. Let $\mathfrak{B}$ be the subsystem of $\mathfrak{S}$ containing all elements of $\mathfrak{T}$ and all elements of $\mathfrak{U}$, and no other elements. Since $\mathfrak{T}_1 + a$ is equivalent to the system $\mathfrak{T}$ contained in $\mathfrak{B}$, one obtains from $\mathfrak{B}$ a system $\mathfrak{B}'$ equivalent to $\mathfrak{B}$ by

replacing α_1 by a . The system $\mathfrak{B}'$ contains $\mathfrak{U} + a$, and since this is equivalent to $\mathfrak{U}$, one obtains a system $\mathfrak{B}''$ equivalent to $\mathfrak{B}'$ by omitting the element a from $\mathfrak{B}'$. Thus $\mathfrak{B}$ would be equivalent to a proper part $\mathfrak{B}''$ of itself, hence reducible. But this is impossible, because $\mathfrak{B}$ is a part of the irreducible system $\mathfrak{S}$.

Thus $\mathfrak{T}$ must indeed be contained in every subsystem of $\mathfrak{S}$ on which a is algebraically dependent. Therefore, if $\mathfrak{S}_i$ ($i = 1, \dots, m$) denotes the system left after omitting α_i from $\mathfrak{S}$, then a is algebraically independent of the irreducible system $\mathfrak{S}_i$, so that $\mathfrak{S}_i + a$ is likewise irreducible. But $\mathfrak{S}_i + a$ arises from $\mathfrak{S}$ by replacing α_i by a , or, what is the same thing, by replacing $\mathfrak{T} = \mathfrak{T}_i + \alpha_i$ by the equivalent system $\mathfrak{T}_i + a$. Hence $\mathfrak{S}$ and $\mathfrak{S}_i + a$ are equivalent systems. Thus: *One obtains an irreducible system equivalent to $\mathfrak{S}$ by replacing one of the elements of $\mathfrak{T}$ by a .* – On the other hand, if γ is an element of $\mathfrak{S}$ not occurring in $\mathfrak{T}$, and if $\mathfrak{S}'$ remains after omitting it, then $\mathfrak{S}'$ contains $\mathfrak{T}$; hence a is algebraically dependent on $\mathfrak{S}'$, and the system $\mathfrak{S}' + a$ arising from $\mathfrak{S}$ by replacing γ by a is reducible. It is equivalent to $\mathfrak{S}'$, and since $\mathfrak{S}'$, as a proper part of the irreducible system $\mathfrak{S}$, cannot be equivalent to $\mathfrak{S}$, neither can $\mathfrak{S}' + a$ be equivalent to $\mathfrak{S}$. This completes the proof of 7 in all parts.

8. *Let $\mathfrak{U}$ and $\mathfrak{B}$ be finite irreducible systems of m and n elements, respectively; suppose $n \leq m$ and that $\mathfrak{B}$ is algebraically dependent on $\mathfrak{U}$. Then, in the case $m = n$, the systems $\mathfrak{U}$ and $\mathfrak{B}$ are equivalent; in the case $n < m$, $\mathfrak{U}$ is equivalent to an irreducible system consisting of $\mathfrak{B}$ and $m - n$ elements from $\mathfrak{U}$.*

Let $u_1, \dots, u_m$ be the elements of $\mathfrak{U}$, and $v_1, \dots, v_n$ those of $\mathfrak{B}$. Since v_1 is transcendental over $\mathfrak{K}$ and algebraically dependent on $\mathfrak{U}$, Theorem 7 gives at least one element of $\mathfrak{U}$ such that replacing it by v_1 changes $\mathfrak{U}$ into an equivalent irreducible system. This proves the assertion for $n = 1$. For $n > 1$ the proof is by complete induction. Let $\mathfrak{B}'$ be the system consisting of $v_1, \dots, v_{n-1}$. Since it is irreducible and algebraically dependent on $\mathfrak{U}$, we may regard it as proved that $\mathfrak{U}$ is equivalent to an irreducible system $\mathfrak{U}'$ consisting of $\mathfrak{B}'$ and $m - n + 1$ elements from $\mathfrak{U}$. The element v_n is algebraically dependent on $\mathfrak{U}'$, but not on $\mathfrak{B}'$. Hence, if $\mathfrak{T}$ is the smallest subsystem of $\mathfrak{U}'$ on which v_n is algebraically dependent, then $\mathfrak{T}$ contains at least one of the $m - n + 1$ elements from $\mathfrak{U}$. Replacing such an element by v_n turns $\mathfrak{U}'$ into an irreducible system equivalent to $\mathfrak{U}$, consisting of $\mathfrak{B}$ and $m - n$ elements of $\mathfrak{U}$. In the case $m = n$, therefore, $\mathfrak{U}$ and $\mathfrak{B}$ are equivalent.

We also see from this that an irreducible system $\mathfrak{B}$ algebraically dependent on $\mathfrak{U}$ cannot contain more than m elements. Otherwise a proper part $\mathfrak{B}'$ of $\mathfrak{B}$ consisting of m elements would, by what has just been proved, be equivalent to $\mathfrak{U}$. Then $\mathfrak{B}$ would be algebraically dependent on $\mathfrak{B}'$, hence reducible. Thus:

9. *An irreducible system $\mathfrak{B}$ algebraically dependent on a finite system $\mathfrak{U}$ cannot contain more elements than that system; in the case of equal numbers of elements the two systems are equivalent.*